Lab 1 – Databricks Files

Lab Summary

In this lab, we learn how to work with **Databricks workspace files**.

We will:

- Create text files inside workspace
- Read and write files using Python
- Create and use Python `.py` files
- Create and read JSON mapping files
- Convert Pandas DataFrame to PySpark DataFrame

This lab helps in collaborative development and managing static/config files inside Databricks.

Prerequisites

- Databricks Workspace access
- Serverless / Cluster running
- Basic Python knowledge
- Basic Pandas & PySpark knowledge

Steps to Complete Lab

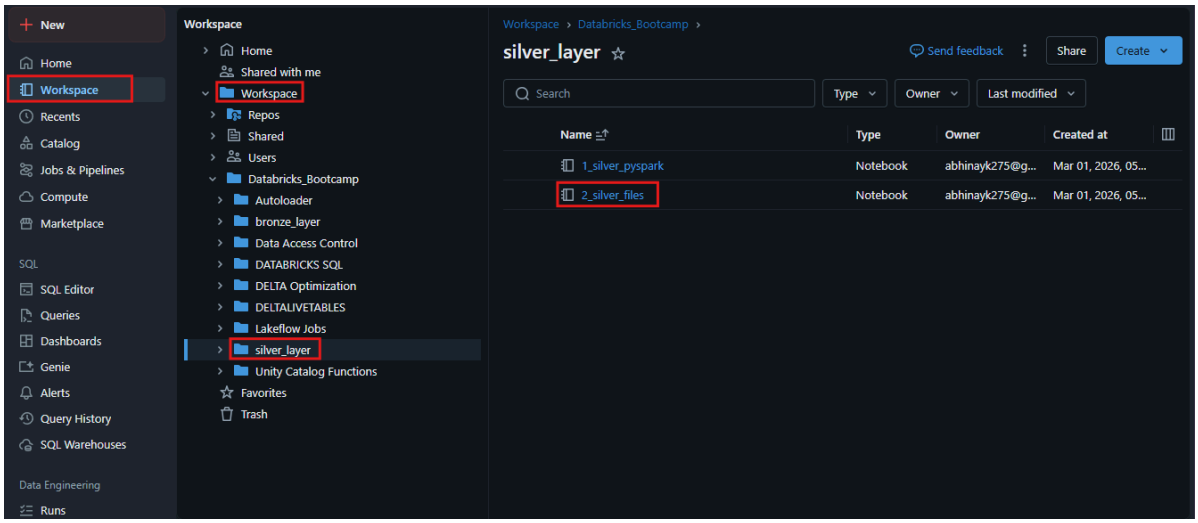
Task 1 – Organize Workspace Structure

1.1 Create Silver Layer Folder

1. Go to **Workspace**
2. Navigate to your project folder (Databricks Bootcamp)
3. Click on silver_layer

4. **Open file**:

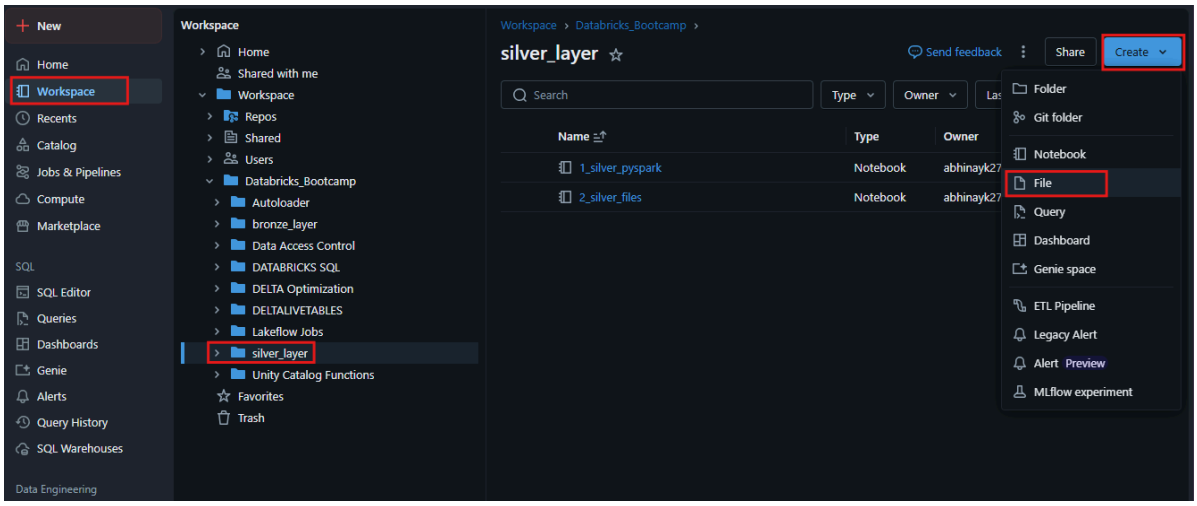
```
2_silver_files
```



Task 2 – Create and Read Text File

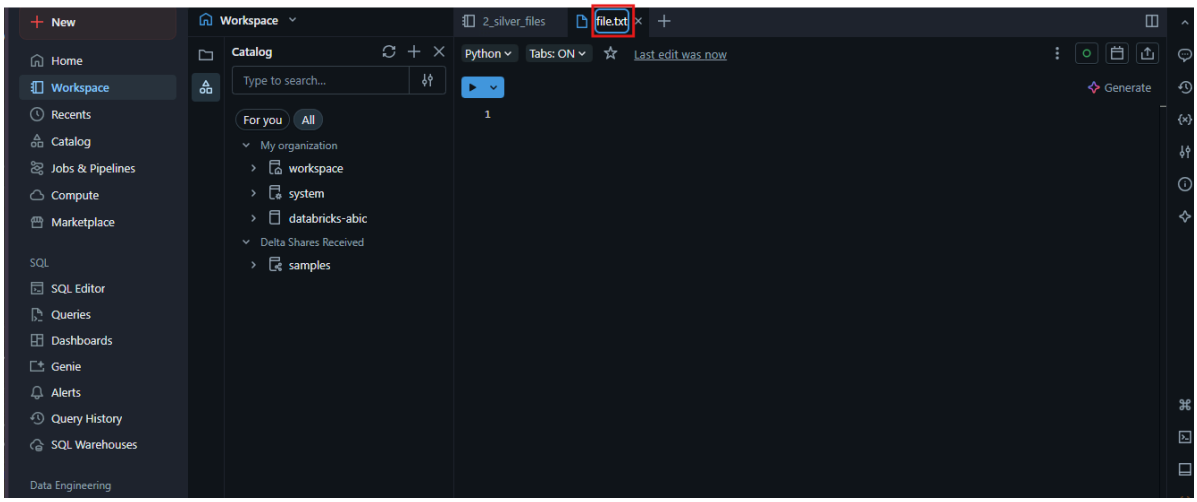
2.1 Create Text File Manually

1. Navigate to your workspace → **Create** → **File**



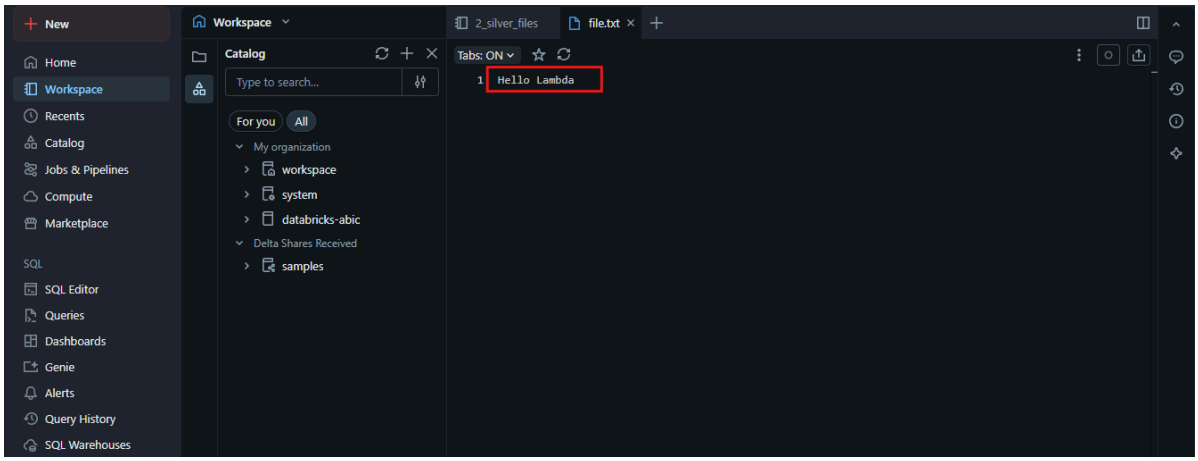
2. Name:

```
file.txt
```



3. Add content:

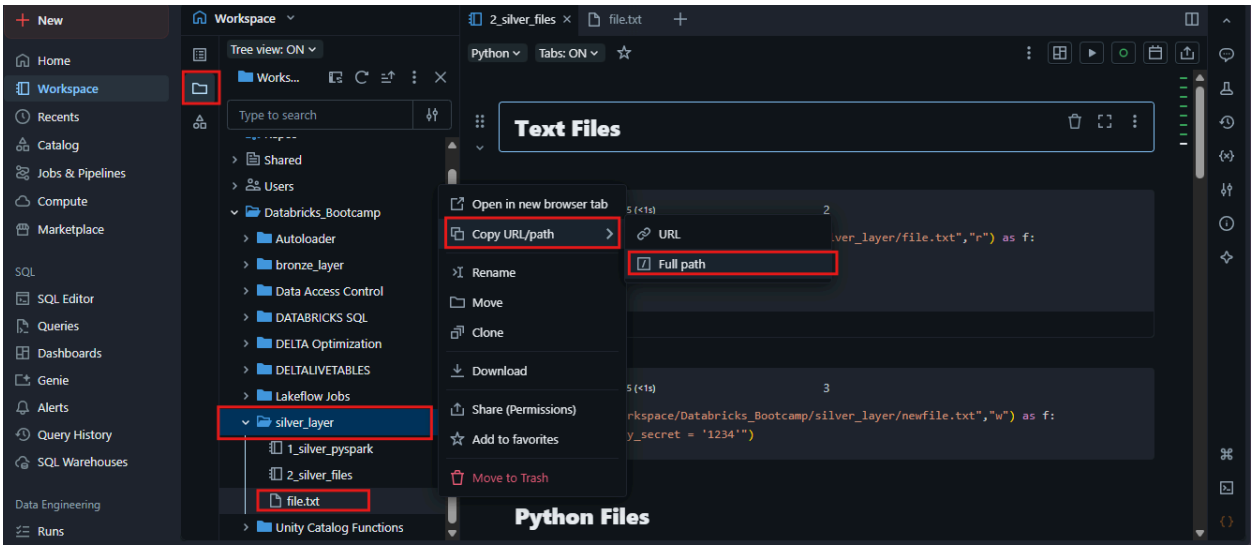
```
Hello Lambda
```



2.2 Read Text File Using Python

Copy file path:

Click → three dots → **Copy full path**

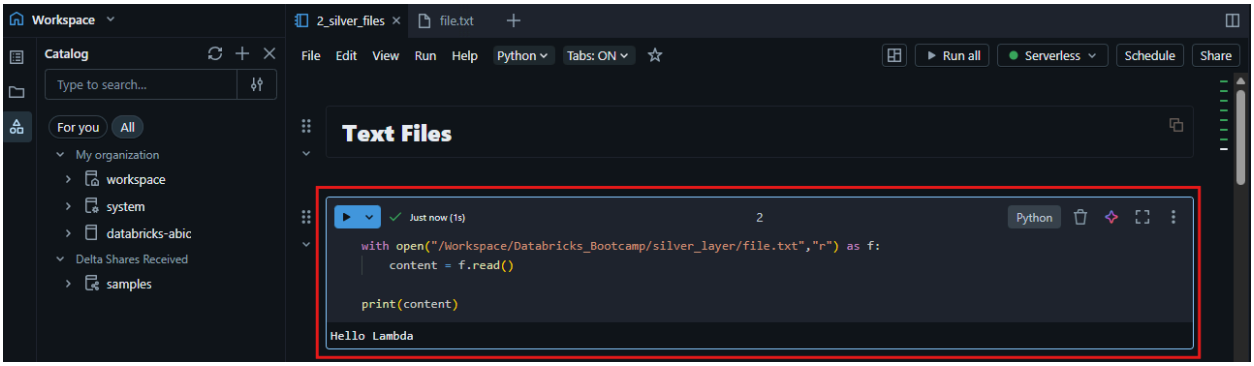


Use Python code to read file:

```
with open("FULL_FILE_PATH", "r") as f:
    content = f.read()

print(content)
```

Output:

```
Hello Lambda
```

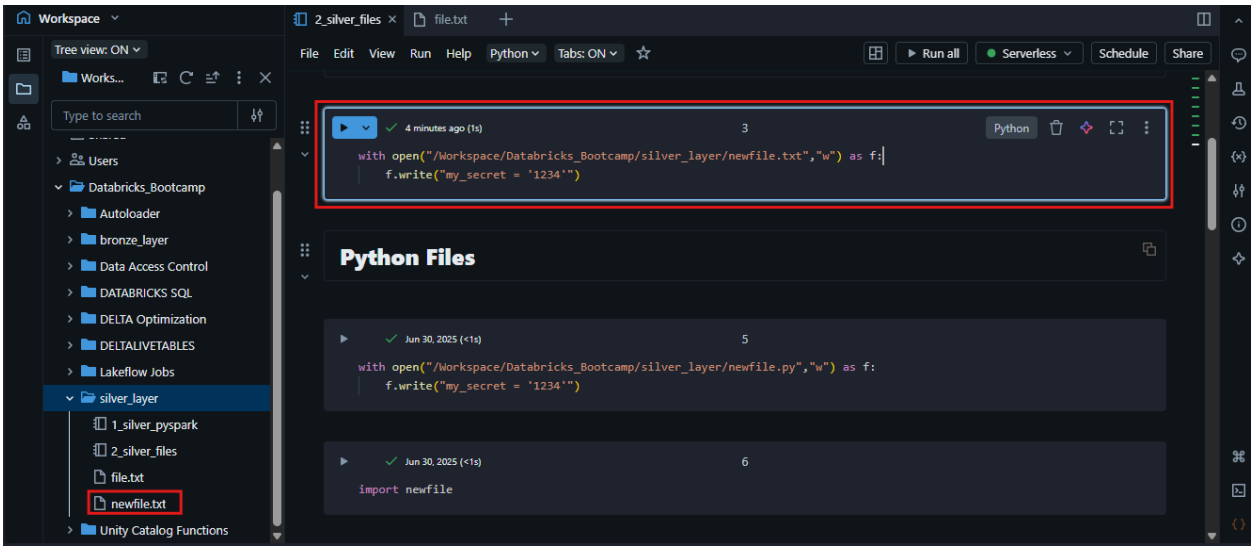


Task 3 – Create File Using Python Code

3.1 Create New File Programmatically

```
with open("FULL_FILE_PATH/new_file.txt", "w") as f:
    f.write("Hello Bro")
```

Refresh workspace → file will appear.

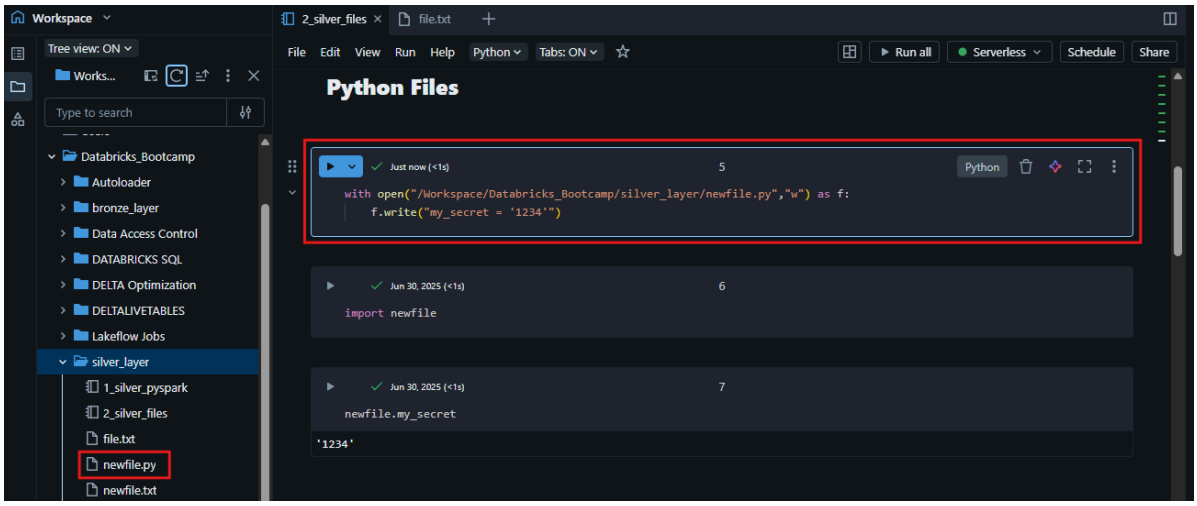


Task 4 – Create and Use Python (.py) File

4.1 Create Python File

1. Create new file: *new_file.py with variable my_secret = 1234*

```
with open("/Workspace/Databricks_Bootcamp/silver_layer/new
file.py","w") as f:
    f.write("my_secret = '1234'")
```

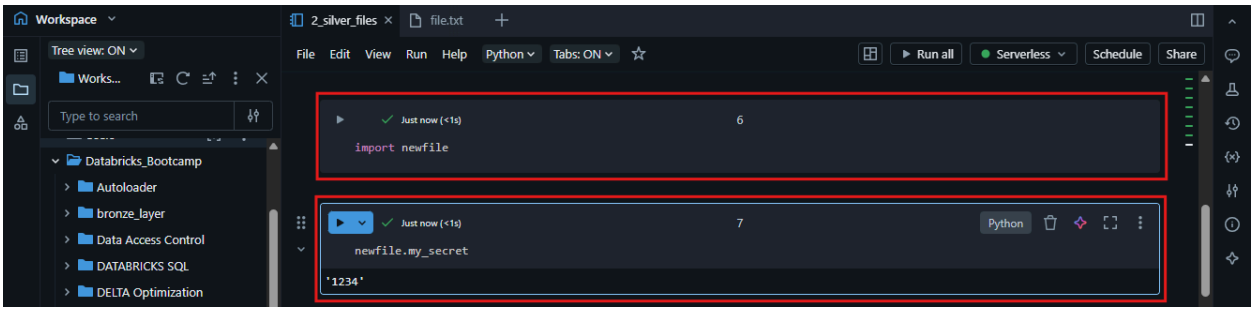


4.2 Import Python File in Notebook

```
import new_file

print(new_file.my_secret)
```

Output:

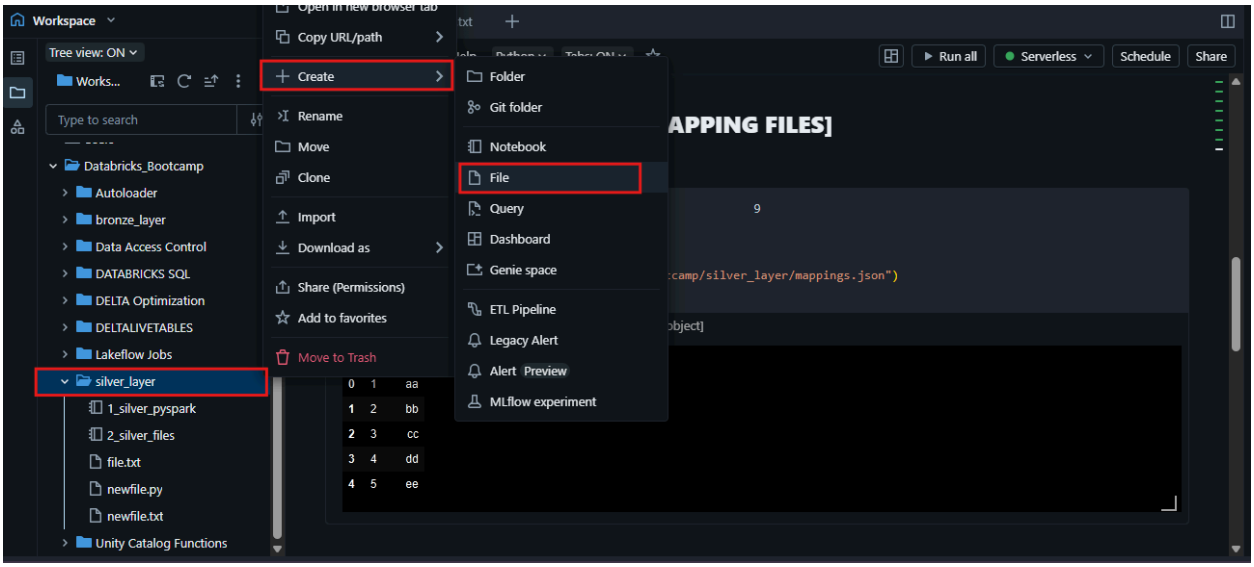
```
1234
```



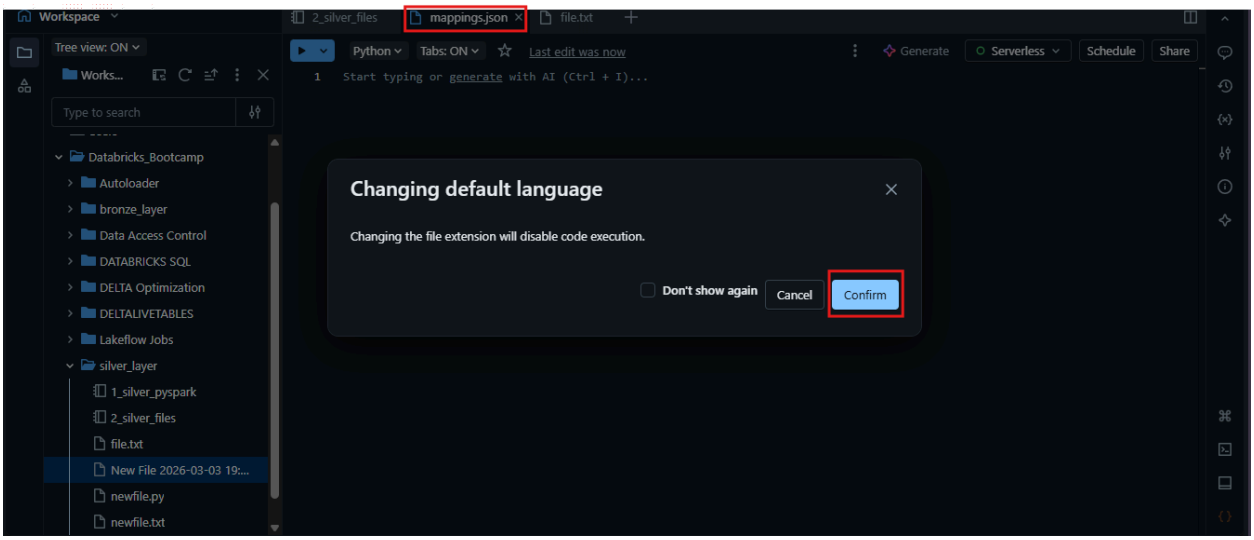
Task 5 – Create and Read JSON Mapping File

5.1 Create JSON File

Create:

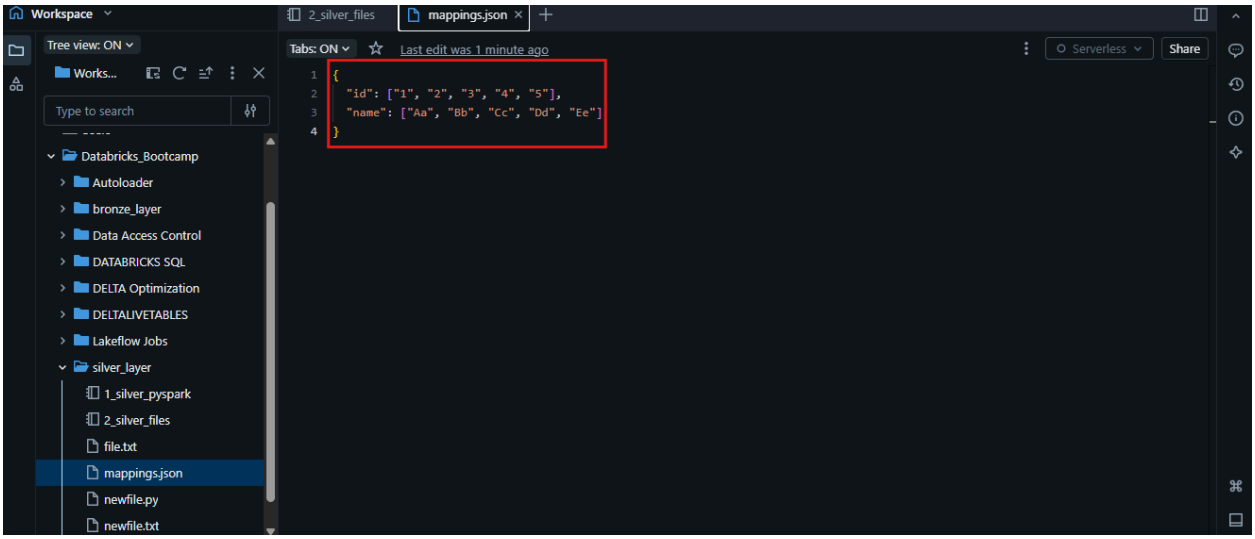



```
mapping.json
```



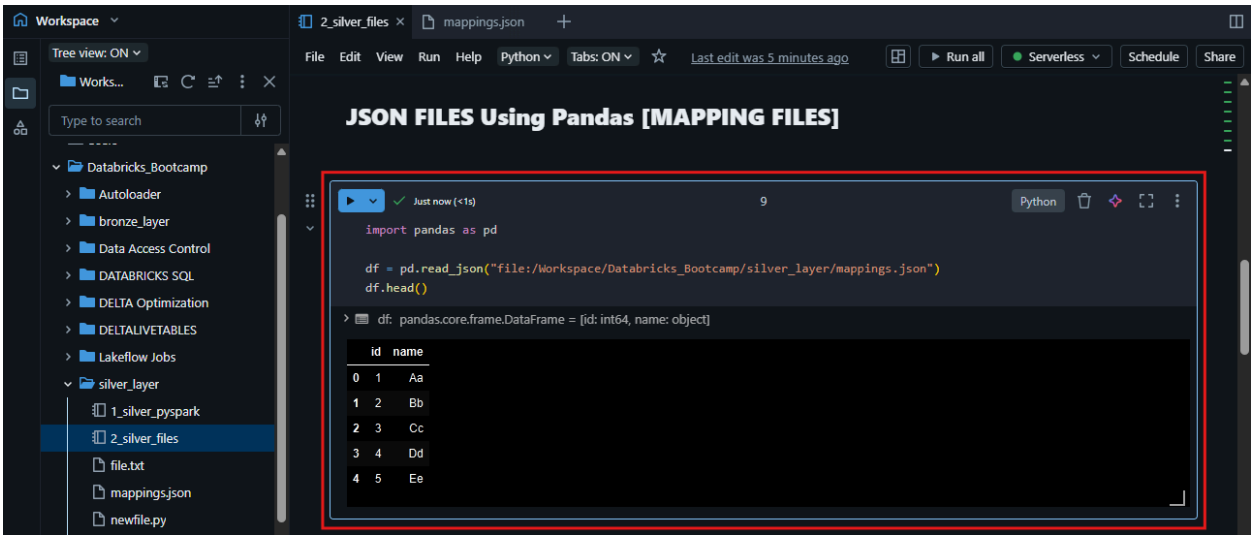
Add:

```
{
  "id": ["1", "2", "3", "4", "5"],
  "name": ["Aa", "Bb", "Cc", "Dd", "Ee"]
}
```



5.2 Read JSON Using Pandas

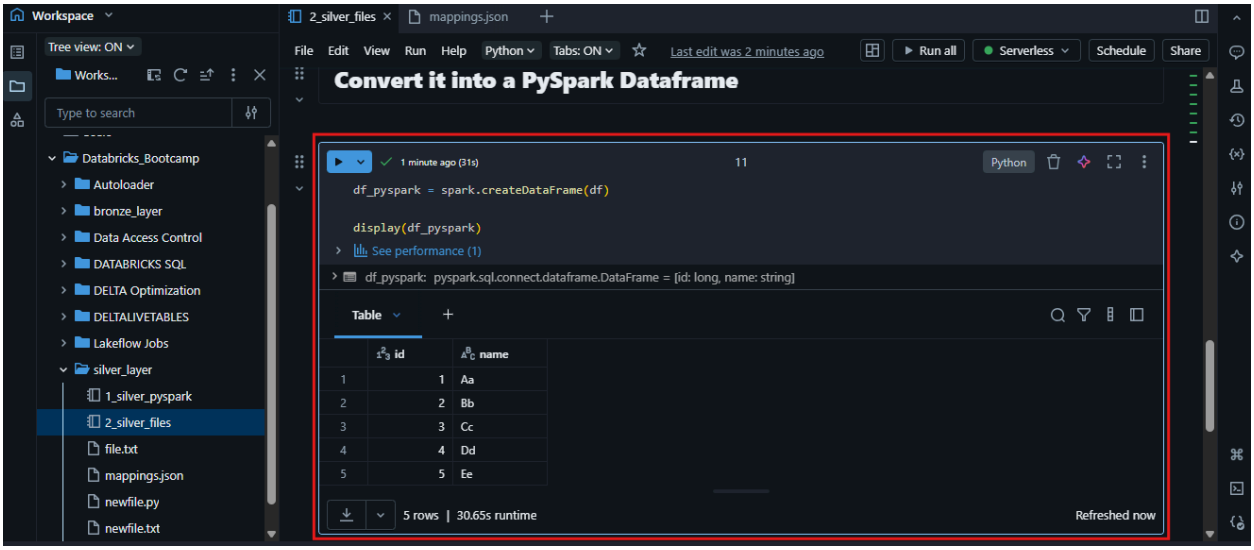
```
import pandas as pd

df = pd.read_json("FULL_PATH/mapping.json")
df.head()
```



5.3 Convert Pandas → PySpark DataFrame

```
df_pyspark = spark.createDataFrame(df)
display(df_pyspark)
```

Now it becomes a PySpark DataFrame.



Important Notes

- Pandas is better for small static files

- PySpark is better for large distributed data
- Always use double quotes in file paths if single quotes cause issues
- Always copy full path from workspace

Outcome

After completing this lab, you can:

- Organize Databricks workspace
- Create and manage files inside workspace
- Read/write files using Python
- Create reusable Python modules
- Read JSON mapping files
- Convert Pandas DataFrame to PySpark DataFrame